PETRA ZLATARIĆ-NOVOSEL

Machine Learning Engineer | NLP & Document AI
Osijek, Croatia

PROFESSIONAL SUMMARY

Machine learning and data science professional with 5+ years of experience across machine learning and data science, including production NLP systems. Experienced across the model lifecycle, from text dataset preparation and experimentation to deployment and production monitoring. Doctoral researcher in named entity recognition for Croatian legal texts, with hands-on Azure deployment experience and a personal RAG prototype.

EXPERIENCE

ML Engineer | Slavonica Document AI d.o.o. | 03/2023–Present

- Built a classifier for scanned incoming public administration documents, achieving 94% accuracy and 0.91 macro F1 across 23 classes; served approximately 15,000 documents per day.
- Fine-tuned XLM-R and BERTić transformer models for document classification and named entity recognition in Croatian administrative texts.
- Deployed models as FastAPI services in Docker on Azure App Service; used Azure ML and MLflow for experiment tracking and model registry.
- Built an annotated dataset of 4,000 Croatian administrative documents, with annotation guidelines and two annotators; achieved inter-annotator kappa of 0.87.
- Implemented weekly production drift monitoring for input text length, vocabulary, and predicted class distribution, with alerts at PSI > 0.2. Detected a silent OCR change that reduced accuracy by 9 percentage points; the issue was fixed within a week.
- Mentor two students on work placements.

Data Scientist | Baranjska kreditna banka d.d. | 09/2021–02/2023

- Developed customer churn prediction models using gradient boosting and LightGBM for the sales team.
- Classified customer complaint text into 11 topics, progressing from a TF-IDF and logistic regression baseline to fastText; results supported weekly reporting to customer service management.
- Produced SQL reports using the Oracle data warehouse.

Research Project Assistant | Faculty of Applied Informatics, University of Drava | 10/2020–08/2021

- Built a corpus of Croatian legal texts and prepared annotations; wrote supporting Python scripts.

SELECTED PROJECT

Retrieval-Augmented Question Answering Demo | Personal project | 2025

- Built a weekend prototype for question answering over Croatian laws using sentence-transformer embeddings, FAISS, and an open-weight LLM accessed through an API; checked results manually. Not used in production.

EDUCATION

Doctoral Studies in Computer Science | University of Drava | 2022–Present
Expected 2027. Research topic: named entity recognition in Croatian legal texts.

Master of Engineering in Computer Science | University of Drava | 2020

TECHNICAL SKILLS

Python, PyTorch, Hugging Face Transformers, scikit-learn, LightGBM, fastText, spaCy, pandas, NumPy, SQL, MLflow, FastAPI, Docker, Azure ML, Azure App Service, Azure Functions, Evidently, PostgreSQL, FAISS, sentence-transformers, Git, Linux

CERTIFICATION

Microsoft Certified: Azure AI Engineer Associate | 2024

PUBLICATION

Zlatarić-Novosel, P. et al. (2024). "Named entity recognition for Croatian administrative documents." Regional workshop on Slavic language processing.

LANGUAGES

Croatian (native), English (C1), German (B2)